MILITARY SPECIFICATION

ARSENIC PENTOXIDE, ANALYZED REAGENT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one analyzed reagent grade of arsenic pentoxide (As_2O_5).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-W-572	-	Wood-Preservative; Water-Repellent.
PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Box, Fiberboard.
PPP-C-186	-	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.
PPP-C-301	-	Chemicals, Dry and Paste, Packaging and Packing of.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-P-15011	-	Pallets, Material Handling, Wood, Post Construction, 4-Way Entry.

FSC 6810

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:56Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" x 48" 4-Way (Partial) Pallet Skids, Runners, or Pallet-Type Base.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should cite "49 CFR 171-179.")

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1193-66 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

3.1 Chemical characteristics. Arsenic pentoxide shall conform to the chemical characteristics of table I when tested as specified therein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:56Z
Check the source to verify that this is the current version before use.

Table I. Chemical characteristics

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Assay (as As_2O_5)	99.0	--	4.2.4.1
Arsenic trioxide (As_2O_3)	--	0.05	4.2.4.2
Chloride (Cl)	--	0.005	4.2.4.3
Heavy metals (as Pb)	--	0.010	4.2.4.4
Iron (Fe)	--	0.010	4.2.4.5
Nitrate (NO_3)	--	To pass test	4.2.4.6

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the arsenic pentoxide produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of preparation for delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Sampling for test shall be conducted in accordance with table II. A representative specimen of approximately 1/4 pound shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:56Z
Check the source to verify that this is the current version before use.

Table II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2-25	2
26-150	3
151-1,200	5
1,201-7,000	8
7,001-20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The sample unit shall be one filled and closed unit, intermediate, or shipping container, as applicable, ready for shipment. Sample containers and the preparation for delivery thereof shall be examined for the following defects using an AQL of 4.0 percent defective:

(a) Contents per container not as specified.
(b) Container not as specified.
(c) Container closure not as specified.
(d) Container damaged or leaking.
(e) Fiberboard fillers and pads, strapping, or tape not as specified.
(f) Wood preservation missing or incorrect (level A packing only).
(g) Marking incorrect, missing, or illegible.

4.2.3.2 For test. Approximately equal portions of all of the specimens taken in 4.2.2.2 shall be thoroughly mixed to form a composited specimen of no less than 1/4 pound. The composited specimen shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Assay (as As_2O_5). Weigh to the nearest 0.1 milligram (mg) approximately 0.25 gram (g) of the specimen into a small iodine flask. Add 50 milliliters (ml) of water, heat to near boiling with stirring, and add 10 ml of 1N sodium hydroxide solution. When solution is complete, neutralize with 10 ml of 1N hydrochloric acid. Heat the solu-

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:56Z
Check the source to verify that this is the current version before use.

tion to 50° ± 2° C, add 5 g of potassium iodide, stopper the flask, and allow to stand in the dark for 30 minutes. Titrate the liberated iodine with 0.1N sodium thiosulfate solution, adding starch indicator solution just before the end point. Calculate the percent arsenic pentoxide as follows:

$$\text{Percent arsenic pentoxide} = \frac{5.75\ AB}{W}$$

where: A = Milliliters of sodium thiosulfate solution used,
B = Normality of sodium thiosulfate solution, and
W = Weight of specimen in grams.

4.2.4.2 Arsenic trioxide (As_2O_3). Weigh to the nearest 0.1 mg approximately 5 g of the specimen and mix with 50 ml of water. Add 10-percent sodium hydroxide solution until the solution is slightly alkaline to litmus paper. Add 2 drops of concentrated hydrochloric acid and then dissolve 1 g of sodium bicarbonate into the solution. Add 5 ml of starch indicator solution and titrate with 0.1N iodine solution. Calculate the percent arsenic trioxide as follows:

$$\text{Percent arsenic trioxide} = \frac{4.95\ AB}{W}$$

where: A = Milliliters of iodine solution used,
B = Normality of iodine solution, and
W = Weight of specimen in grams.

4.2.4.3 Chloride. Weigh 0.400 g of the specimen and dissolve in 1 ml of concentrated nitric acid and 20 ml of water. Add 1 ml of 1.7-percent silver nitrate solution. Any turbidity produced shall be no greater than that produced in a blank to which 0.02 mg of chloride ion has been added.

4.2.4.4 Heavy metals (as Pb). Weigh 0.200 g of the specimen and dissolve in 5 ml of hot water in a small porcelain crucible. Add 10 mg of sodium carbonate and 5 ml of concentrated hydrobromic acid and evaporate to dryness under a well-ventilated hood. Re-evaporate the residue with 5 ml of water and 5 ml of concentrated hydrobromic acid. Warm the residue with 0.5 ml of 1N hydrochloric acid, dilute to 40 ml with water, and add 10 ml of saturated hydrogen sulfide water. Any color produced shall be no darker than that produced in a control containing 0.02 mg of lead ion and the quantities of reagents used in the specimen test in the same final volume.

4.2.4.5 Iron. Weigh 0.500 g of the specimen and dissolve in 5 ml of hot water in a small porcelain dish. Add 10 ml of hydrobromic acid and evaporate to dryness under a hood. Warm the residue with 2 ml of concentrated hydrochloric acid until dissolved, dilute with 25 ml of

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:56Z
Check the source to verify that this is the current version before use.

water, saturate the solution with hydrogen sulfide, filter, and wash with 10 ml of water containing 3 drops of concentrated sulfuric acid and some hydrogen sulfide. Evaporate the filtrate to dryness and ignite. To the filtrate residue, add 2 ml of concentrated hydrochloric acid, 2 ml of water, and a few drops of concentrated nitric acid. Slowly evaporate to dryness on a steam bath. Take up with 1 ml of concentrated hydrochloric acid and water to make 50 ml of solution. To 20 ml of the solution, add 2 ml of concentrated hydrochloric acid and dilute to 50 ml with water. Add 30 mg of ammonium persulfate and 3 ml of 25-percent ammonium thiocyanate solution and mix. Any red color produced shall be no darker than that produced in a control containing 0.02 mg of iron ion and the quantities of reagents used in the specimen test in the same final volume.

4.2.4.6 Nitrate.

(a) Indigo carmine solution. Dissolve 0.1 g of indigo carmine which has been dried for 2 hours at 105° to 110° C in a mixture of 10 ml of concentrated sulfuric acid and 80 ml of water and dilute to 100 ml with water. Keep the solution protected from light. Discard after 30 days.

(b) Procedure. Weigh 3.00 g of the specimen and mix with 10 ml of water. Add 0.25 ml of indigo carmine solution prepared as specified in (a) and 10 ml of concentrated sulfuric acid. The blue color produced shall not be completely discharged after 5 minutes.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. One-quarter pound + 0.05 ounce of arsenic pentoxide shall be packaged in accordance with the applicable level A packaging requirements of PPP-C-301 in a type I, class 1 glass bottle, in inclosure b as specified therein. The bottle shall also conform to glass type b, light penetration grade optional, with outer seal A band or tape of PPP-C-186.

5.1.1.1 Intermediate packaging. Twelve bottles of arsenic pentoxide, packaged as specified in 5.1.1, shall be intermediately packaged in accordance with the applicable requirements of PPP-C-301 except that table I therein shall not be used. The bottles shall be arranged in a three bottle by four bottle pattern. Fiberboard separators shall not be required.

5.1.2 Level C. One-quarter pound of arsenic pentoxide shall be packaged in a glass bottle as specified in 5.1.1 except that inclosure b of PPP-C-301 shall not be required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:56Z
Check the source to verify that this is the current version before use.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Four intermediate packages, prepared as specified in 5.1.1, shall be packed upright in a nailed wood, cleated plywood, or wirebound wood box as specified (see 6.2). The nailed wood box shall conform to class 2, style 4-1/2, for a type 2 load of PPP-B-621. The cleated plywood box shall conform to type overseas, style I or J, for a type 2 load of PPP-B-601. The wirebound box shall conform to class 3, style optional, for a type 2 load of PPP-B-585. The use of fiberboard fillers and pads shall be in accordance with PPP-C-301 to assure a tight pack. Each container shall be closed and strapped using galvanized strapping as specified in the appendix to the applicable box specification. In addition, unless specific provisions are made for preservation in the applicable box specification, each box or the finished, unpainted wood parts thereof shall be completely immersed for a minimum of 3 minutes in a water-repellent wood preservative conforming to composition A of TT-W-572.

5.2.2 Level B. Arsenic pentoxide shall be packed as specified in 5.2.1 except that the box shall conform to style RSC, grade V3c of PPP-B-636. The box shall be closed in accordance with the requirements for taping style RSC boxes as exterior containers, and then reinforced by tape banding as specified in the appendix to PPP-B-636. No preservative shall be required.

5.2.3 Level C. Twelve bottles of arsenic pentoxide, packaged as specified in 5.1.2, shall be packed in a fiberboard box conforming to style RSC, type CF, class domestic, variety SW, grade 125 of PPP-B-636. The box shall be fitted with fiberboard half-slotted separators which shall form close fitting cells for each of the bottles and which shall be the same height as the bottles. All inner faces of each box shall be lined with full box sized fiberboard pads. Additional pads shall be added where needed to assure a tight fit. Fiberboard pads and separators shall conform to type CF, class domestic, variety SW, grade 125 of PPP-F-320. The box shall be closed in accordance with the requirements for taping as specified in the appendix to PPP-B-636.

5.3 Marking. In addition to the marking specified for unit containers in 5.3.1 and any marking required by the contract or order (see 6.2), each container shall be marked in accordance with Department of Transportation (DOT) regulations ("POISON" label) and MIL-STD-129.

5.3.1 Special marking. Each unit container shall be durably and legibly labeled to show the following information:

(a) Empirical formula.
(b) Formula weight.
(c) Manufacturer's lot number.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:56Z
Check the source to verify that this is the current version before use.

(d) Assay.
(e) Analysis or maximum limits of impurities.
(f) Net weight of contents.
(g) Precautionary marking:

DANGER! MAY BE FATAL IF SWALLOWED

Avoid breathing dust or spray mist.
Avoid contact with skin, eyes, and clothing.
Wash thoroughly after handling.

POISON

Call a physician.
First aid if swallowed: If conscious, make patient vomit immediately by putting finger down throat or by giving warm mustard water or salt water freely.

5.4 Palletization. When specified in the contract or order (see 6.2), shipping containers shall be palletized in accordance with the applicable requirements of MIL-STD-147 using load type I except that strapping shall be restricted to galvanized steel of 1.25-inch minimum width and 0.035-inch minimum thickness and the load tie-down straps parallel to the 48-inch pallet dimension shall be modified to serve as load unitizing straps, containing the load but not the pallet. One load tie-down strap shall be added to, and centered between, those straps which are normally required to be parallel to the 40-inch pallet width. One horizontal strap shall be required for each tier of containers. The pallet shall conform to style 1, type optional, class 1, preserved of MIL-P-15011. All wood storage aids, if used, shall be preserved as specified for the pallet.

6. NOTES

6.1 Intended use. Arsenic pentoxide is intended for use as a laboratory reagent.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required (see 5.1 and 5.2).
(c) Kind of box required for level A packing (see 5.2.1).
(d) Additional marking, if required (see 5.3).
(e) Palletization, if required (see 5.4).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:56Z
Check the source to verify that this is the current version before use.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

6.5 Supersession data. This specification includes the requirements of Edgewood Arsenal Purchase Description 4-131, dated 19 November 1952.

Custodians:

Army - MU
Air Force - 68

Review activities:

Army - MD, MU(FA)
Air Force - 68
DSA - GS

User activities:

Army - MI, SM

Preparing activity:

Army - MU(EA)

Project No. 6810-0466

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:56Z
Check the source to verify that this is the current version before use.

TO DETACH FORM - CUT ON THIS LINE

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:56Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding Officer
Edgewood Arsenal
ATTN: SMUEA-QAES
Edgewood Arsenal, Maryland 21010

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:56Z
Check the source to verify that this is the current version before use.